

Lab Experiment-VII: Major and Minor Losses

Objectives

- (i) Measure Major losses in a straight tube for various flow rates.
- (ii) Measure Minor losses in bends, valves, contractions etc., for various flow rates.
- (iii) Quantify the loss coefficient in various minor loss components.

The Essential Background

1. What are Major and Minor head losses in a flow? Why is it important to quantify these losses?
2. How does the major and minor losses depend on the flow rate?
3. What is *Friction coefficient*? How is *Minor loss coefficient* defined?
4. How are head losses quantified in Valves, contractions and expansions, pipe bends and elbows?
5. What does a Moody's chart do?
6. What is the *kinetic energy coefficient*?

Task 7.1: In the report for Lab Exp.-VII , concise answers to all the above questions have to be submitted.



Figure 7: Set up for measuring major and minor losses in a flow network.

About the experiment

The experimental set up for the measurement of the Major and Minor losses has been shown in figure 7. This is a simple set up where liquid water is flown through moderately large lab-scale network, where loss inducing components such as pipes, bends, elbows, valves etc. are present in series. The water is pumped from a reservoir through this network, and then back to the reservoir again. Manometers may be used in the appropriate pressure tapings to measure the head losses incurred in the various components in the network.

Task 7.2:

- (a) Measure the head losses in the various components in the network for at least 4 different flow rates. Repeat your measurement for the same flow rate multiple times. Some of the flow rate you choose should be sufficiently small so that the flow remains laminar, while other choices should result in turbulent flows.
- (b) Calculate the loss coefficients, and from there compute the equivalent length in each of the minor loss inducing component. If possible, compare them with the known values from the literature. To this end, you may consult your textbook.
- (c) You should measure the major losses in the three straight tubes, made of three different materials (what are these materials?) and report how does the choice of the tube material impact the losses?
- (d) Quantify and report experimental uncertainties and errors.

Tasks 7.1 and 7.2 should be included in the report for this experiment.